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Course: Computer vision

Analytical problem solving

1)

a) Purpose of Gray level transformation

Gray level transformation is used to improve the appearance of an image by changing the intensity value of pixels.

- ⇒ Enhancing image Contrast
- ⇒ Highlighting important features
- ⇒ Making dark areas brighter or bright areas darker
- ⇒ Improving image quality for further processing

b) Formula

$$S = 255 - x$$

Original pixel	Calculation	Transformed pixel
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Transformed pixel values:

205, 153, 105, 55

Interpretation

- ⇒ Dark pixel becomes brighter
- ⇒ Bright pixel becomes Dark
- ⇒ The image appears like a dark

2)
a) Histogram

Histogram equalization Improves image Contrast by spreading pixel Intensity values over the entire available range

Advantages

Enhance low-Contrast images

Reveals hidden details

Makes brightness distribution more uniform

b) Compute CDF

Intensity	frequency
0	4
1	6
2	10
3	6

Gaussian Smoothing

Reduce image noise effectively
Produce smoother images

b)

kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$\begin{aligned} &= (1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + \\ &\quad (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10) \\ &= 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10 \\ &= 360 \end{aligned}$$

Interpretation

$$\frac{360}{16}$$

$$= 22.5$$

⇒ Noise is reduced

⇒ Image becomes smoother

Total pixels
 $N = 4 + 6 + 10 + 6 = 26$

Intensity	Frequency	Cumulative frequency	CDF
0	4	4	$4/26 = 0.154$
1	6	10	$10/26 = 0.385$
2	10	20	$20/26 = 0.769$
3	6	26	$26/26 = 1.000$

c) Equalized Intensities

Since it is a 4 bit image

Max Intensity

$$L-1 = 15$$

Formula

$$s = (L-1) \times CDF$$

Intensity	CDF	Equalized
0	0.154	$15 \times 0.154 = 2$
1	0.385	$15 \times 0.385 = 6$
2	0.769	$15 \times 0.769 = 12$
3	1.000	$15 \times 1 = 15$

Equalized Intensities:

0 $\rightarrow$ 2

1 $\rightarrow$ 6

2 $\rightarrow$ 12

3 $\rightarrow$ 15

Interpretation

- $\Rightarrow$ pixel values spread across the full intensity range
- $\Rightarrow$ Image Contrast

Edge detection identifies sudden intensity change in image.

Shape recognition
feature extraction

b)

$$\begin{bmatrix} -1 & 2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$\begin{aligned} &= (-1 \times 10) + (-2 \times 10) + (-1 \times 10) \\ &= -40 \\ &= (1 \times 30) + (2 \times 30) + (1 \times 30) \\ &= 120 \end{aligned}$$

$$\begin{aligned} G &= -40 + 120 \\ &= 80 \end{aligned}$$

Interpretation

- ⇒ A strong horizontal edge is detected
- ⇒ Large gradient value indicate stronger edges.

5) a) Increase image Contrast
uses the full grayscale range (0-255)

$$b) \quad s = \frac{(r - r_{\min})}{(r_{\max} - r_{\min})} \times 255$$

$$r_{\min} = 50 \quad r_{\max} = 150$$

$$r = 50$$

$$s = \frac{50 - 50}{150 - 50} \times 255$$

$$= 0$$

$$r = 100$$

$$s = \frac{100 - 50}{150 - 50} \times 255$$

$$r = 150$$

$$= 128$$

$$s = \frac{150 - 50}{150 - 50} \times 255$$

$$= 255$$

Original
50
100
150

Transformed

0

128

255

Interpretation

⇒ The Original range 50-150 is stretched to 0-255

⇒ Image Contrast increases significantly